

PAPER_948: Production Scaling V9 Benchmark (400k calc/s)

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** production_scaling_v9.py (ProductionScalingV9) **Calculator:** ProductionScalingV9BenchmarkCalc (CP4 #532) **CVW:** v2.0.0 compliant

Abstract

We present the v9 production scaling benchmark targeting 400,000 calculations per second, a 14% increase over the v8 target of 350,000. The benchmark suite comprises 12 kernels: the 10 v8 kernels plus two new additions - Centaurus A jet power evaluation and SMBH merger strain damping computation. Performance is measured in calculations per second with automated pass/fail against the 400k target.

1. Benchmark Kernels

Kernel	Operation	Source
1-10	v8 base kernels	production_scaling_v8.py
11	CenA Jet P_BZ	blazar_jet_power_curves_extended.py
12	SMBH Merger D_total	smbh_binary_mergers.py

Target

$$\bar{R} = \frac{1}{K} \sum_{k=1}^K R_k \geq 400,000 \text{ calc/s}$$

where $K = 12$ is the total kernel count.

2. Scaling History

Version	Target (calc/s)	Kernels	Date
v4	100,000	4	Dec 2025
v5	150,000	5	Jan 2026
v6	200,000	6	Feb 2026
v7	300,000	8	Mar 2026
v8	350,000	10	Apr 2026
v9	400,000	12	Apr 2026

3. New Kernels

Kernel 11 - CenA Jet: Evaluates P_{BZ} for Centaurus A at three linewidths. Single-pass complexity $O(1)$.

Kernel 12 - SMBH Merger Strain: Computes $D_{\text{total}}(q) = 0.333 + 0.197(1 - q)$ and applies to fiducial strain. Single-pass complexity $O(1)$.

4. Source Data

- **File:** production_scaling_v9.py
 - **Session:** 213
 - **CP4 Class:** ProductionScalingV9BenchmarkCalc (#532)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Production benchmark history: v4 (Session 191) through v9 (Session 213)